Kalman Filter Theory and Applications
State Space Concepts

https://github.com/musicarroll/kalman_course

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Part III

State Space Concepts

Topics

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State Space Concepts

Topics

- Understanding the Equations: Heuristic Introduction

Part III

State Space Concepts

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- Understanding the Equations: Heuristic Introduction
- Equation Drilldown: Taking the Equations Apart

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- **State Space Concepts** ($\geq$8 videos)

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State Space Concepts

Topics

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- Dynamic Systems: The Concept of State Space

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- **Deriving State Models from Differential Equations**

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State Space Representation

Deriving State Equations from Differential Equations

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- This leads to using a operator polynomial to express an nth-order homogeneous differential equation with constant coefficients:

$$p(D)y = D^n y + a_{n-1}D^{n-1}y + \cdots + a_1 Dy + a_0 y = 0, \quad (1)$$

where we have put the operator polynomial in normal form, such that its leading coefficient is 1 (e.g., dividing through by m in $F = ma$)

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- We now define new variables to replace the derivatives by setting $x_1 = y, x_2 = Dy, \ldots$:

$$
\begin{aligned}
x_1(t) &:= D^0 y(t) := y(t) \\
x_2(t) &:= Dy(t) \\
&\ldots \\
x_n(t) &:= D^{n-1} y(t)
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- Taking the derivative of these n state variables and making use of the original differential equation (1) yields (and using the fact that $Dx_n = D^n y$):

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Dx_1(t) &= x_2(t) \\
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- To obtain the original output y as a function of the internal state of the system, we require one additional vector-matrix equation:

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